

School SaaS Platform — Build Roadmap v1.0

Django 5 + DRF + django-tenants | React + Vite +
Tailwind Kenya CBC Schools (PP1–Grade 12) |
Solo Build | 12 Weeks

Build Phases

Phase 1	Wk 1–2	Foundation	Multi-tenancy, Auth, JWT, React scaffold, Login
Phase 2	Wk 3–4	Students Module	Student records, Admissions, Guardians, CBC classes
Phase 3	Wk 5–6	Finance Module	Fee structures, M-Pesa STK Push, Receipts, Bursar dashboard
Phase 4	Wk 7	Academics & CBC	CBC grading (EE/ME/AE/BE), Attendance, Teacher dashboard
Phase 5	Wk 8	SMS & Comms	Africa's Talking, Fee reminders, Attendance alerts
Phase 6	Wk 9–10	White-label & UX	School branding, Superadmin, Onboarding, Demo seed
Phase 7	Wk 11	Polish	Mobile responsive, Empty states, Skeletons, Testing
Phase 8	Wk 12	Deploy & Ship	Hetzner VPS, Nginx wildcard SSL, CI/CD, live demo

Week-by-Week Tasks

WEEK 1–2 Foundation		■ DONE
BACKEND ✓ Django 5 + django-tenants configured ✓ PostgreSQL schema-per-tenant ✓ Custom User model (5 roles) ✓ JWT auth: login / refresh / logout / me ✓ School + Domain tenant models ✓ School onboarding API ✓ Celery + Redis + Docker Compose	FRONTEND ✓ React + Vite + Tailwind scaffolded ✓ Tenant-aware Axios + JWT interceptor ✓ Zustand auth store ✓ Login page with validation ✓ Role-based redirect (5 roles) ✓ ProtectedRoute + GuestRoute	
WEEK 3–4 Students Module		■ IN PROGRESS
BACKEND ✓ Classroom model (PP1-Grade 12) ✓ Guardian + Student models ✓ Student serializers (list + detail) ✓ Full CRUD + search + transfer endpoints ■ pip install django-filter → INSTALLED_APPS	FRONTEND ✓ AppShell with collapsible sidebar ✓ Reusable UI component library ✓ Students list page (search + filters) ✓ Admit Student form ✓ Student detail page ■ Fix students/index.js exports ■ Wire AppShell into App.jsx routes	
WEEK 5–6 Finance Module		■ TODO
BACKEND ■ FeeStructure CRUD endpoints ■ FeeInvoice auto-generate per student/term ■ M-Pesa STK Push endpoint ■ M-Pesa callback handler ■ Payment recording + receipt PDF ■ Fee balance per student endpoint	FRONTEND ■ Fee structure setup page ■ Fee invoices list (filter by term/class) ■ M-Pesa STK Push button ■ Payment history table ■ Bursar dashboard with totals ■ Receipt download	

WEEK 7 Academics & CBC Grading		■ TODO
BACKEND ■ Subject model (learning area + teacher) ■ CBCGrade model (EE/ME/AE/BE per strand) ■ AttendanceRecord model ■ Bulk attendance endpoint ■ Grade entry + retrieval endpoints	FRONTEND ■ Attendance capture (class roster + toggles) ■ CBC grade entry form ■ Student grades view ■ Teacher dashboard ■ Attendance history	
WEEK 8 SMS & Communication		■ TODO
BACKEND ■ Africa's Talking service wired up ■ SMSLog model + endpoints ■ Fee reminder Celery task ■ Attendance alert task ■ Celery Beat schedule (8am EAT daily)	FRONTEND ■ SMS log table ■ Announcements page ■ Manual SMS trigger ■ Communication dashboard	
WEEK 9–10 White-label & Multi-school UX		■ TODO
BACKEND ■ School profile update (logo, colors, motto) ■ Superadmin endpoints ■ Plan enforcement (student count limits) ■ Demo data seeder (Kenyan names)	FRONTEND ■ School settings page (logo + color picker) ■ CSS variable injection per tenant ■ Superadmin platform dashboard ■ School onboarding flow ■ Dashboard stats wired to API	
WEEK 11 Polish & Testing		■ TODO
BACKEND ■ pytest-django test suite ■ API error standardization ■ Rate limiting on auth + SMS	FRONTEND ■ Mobile responsive audit ■ Loading skeletons on all tables ■ Empty states on all list pages ■ Toast notifications + error boundaries ■ Accessibility audit	
WEEK 12 Deploy & Demo Prep		■ TODO
BACKEND ■ Hetzner VPS (Ubuntu 22.04) setup ■ Nginx wildcard subdomain routing ■ Certbot wildcard SSL (*.yourapp.co.ke) ■ Gunicorn + systemd ■ GitHub Actions CI/CD ■ PostgreSQL backup cron	FRONTEND ■ Vercel / Nginx frontend deploy ■ Cloudflare DNS wildcard config ■ demo.yourapp.co.ke with seed data ■ Demo credentials card ■ School pitch deck updated	

Key Integrations

Service	Purpose	Week	Credentials in .env
M-Pesa Daraja	STK Push payments + callback	Wk 5–6	MPESA_CONSUMER_KEY, SECRET, SHORTCODE, PASSKEY
Africa's Talking	SMS fee reminders + alerts	Wk 8	AT_API_KEY, AT_USERNAME, AT_SENDER_ID
Cloudflare R2	File storage (logos, photos, PDFs)	Wk 1+	R2_ACCOUNT_ID, ACCESS_KEY, SECRET_KEY, BUCKET
Hetzner VPS	Production server Ubuntu 22.04	Wk 12	SSH key + server IP

Cloudflare DNS	Wildcard subdomain routing	Wk 12	*.yourapp.co.ke → VPS IP
Redis	Celery broker + result backend	Wk 1+	REDIS_URL
PostgreSQL	Multi-schema tenant database	Wk 1+	DATABASE_URL

Monetization

Plan	Students	Price / Month	Notes
Trial (90 days)	Up to 100	FREE	Pilot schools — zero risk, get case studies
Starter	Up to 400	KES 2,500	Small primary schools
Growth	Up to 1,000	KES 7,500	Mid-size schools
Enterprise	Unlimited	KES 15,000	Large / multi-campus

SMS credits sold as add-ons (buy bulk from Africa's Talking, resell with margin). Setup fee waived for 6-month commitment. Never commit .env to Git.